

Efficient Frontier Shift: Pre- vs. Post-COVID Indian Equities

IQF Midterm Project

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Does the efficient frontier for Indian equities shift meaningfully between **pre-COVID (2015–2020)** and **post-COVID (2021–2024)**, and what structural forces drive the shift?

Why it matters

- COVID was a sharp structural break in global markets
- India's equity market saw unprecedented post-pandemic stimulus and sectoral rotation
- A shift in the frontier implies a change in the *fundamental* risk-return trade-off available to investors

Policy & Market Structure

- **Unprecedented stimulus:** RBI cut repo rate from 5.15% to 4.0%; massive OMO and LTRO liquidity injections
- **Digital transformation:** COVID accelerated e-commerce, fintech, remote work — reshaping sectoral weights
- **Retail investor boom:** Demat accounts grew from 4 crore (2019) to 14+ crore (2024); SIP flows surged

India-Specific Factors

- **China+1 strategy:** FII inflows as India became alternative manufacturing hub
- **PLI schemes:** Production-linked incentives reshaping industrial landscape
- **Fastest-growing major economy:** Post-COVID growth trajectory diverged from developed markets

Universe

- 45 stocks from NIFTY 50 (as of Jan 1, 2015)
- Source: Yahoo Finance

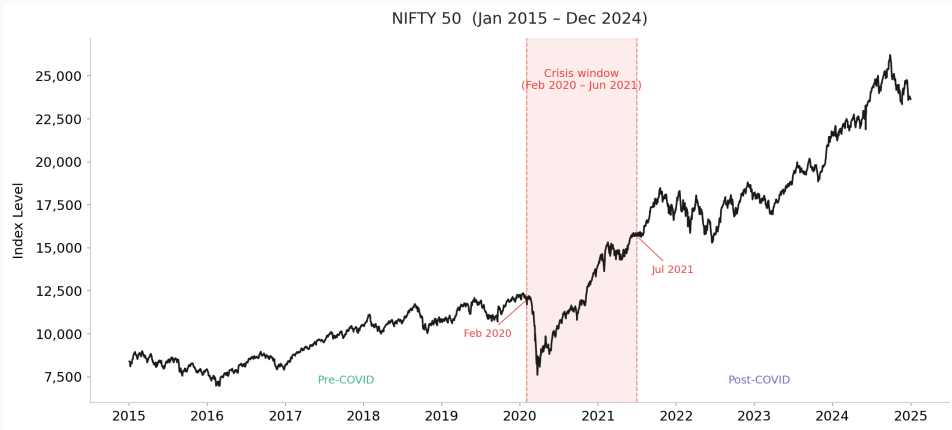
Periods

- Pre-COVID: Jan 2015 – Jan 2020
- Post-COVID: Jul 2021 – Dec 2024
- Analyzed at daily & weekly frequency

Estimation

- Annualized Returns ($\times 252$ daily, $\times 52$ weekly)
- Ledoit-Wolf shrinkage covariance
- Risk-free rate: RBI 91-day T-Bill yield (pre: 6.54%, post: 5.78%)

NIFTY 50 Index: 2015–2024



- NIFTY fell $\approx 26\%$ peak-to-trough (Feb–Mar 2020); the crisis window (shaded) is *excluded* from both estimation periods
- Post-COVID window captures a clean bull-market regime, free of crash-induced distortions

1. **Compute statistics** — annualized returns; Ledoit-Wolf shrinkage applied to Σ for each period
2. **Trace the efficient frontier** — Solve the minimization problem for multiple points to trace the efficient frontier

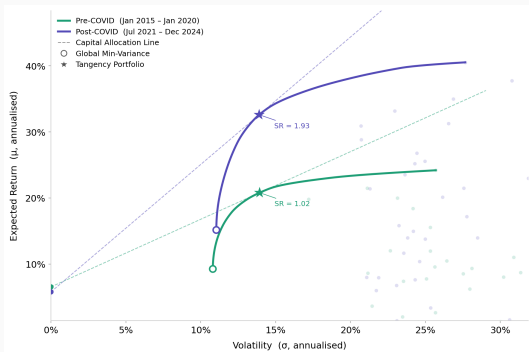
$$\min_{\mathbf{w}} \mathbf{w}^T \Sigma \mathbf{w} \quad \text{s.t.} \quad \mathbf{w}^T \boldsymbol{\mu} = \mu^*, \mathbf{w}^T \mathbf{1} = 1, \mathbf{w} \geq 0$$

sweeping μ^* from GMV to the highest-return asset

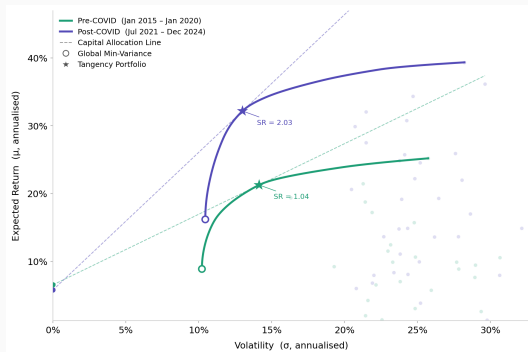
3. **Tangency portfolio** — maximize Sharpe ratio with RBI T-bill as r_f ; compare pre vs. post weights

Efficient Frontier — Daily & Weekly

Daily



Weekly



The post-COVID frontier lies *above and to the right* in both cases — the risk-return trade-off improved materially.

Interpreting the Frontier Shift

What Shifted?

- **Northeast movement:** Higher returns **and** higher volatility
- Return gain (+5 pp) **exceeds** volatility gain (+3 pp)
- **Steeper CAL:** Sharpe ratio improved +6–7% across frequencies
- **Entire frontier** shifted, not just tangency point

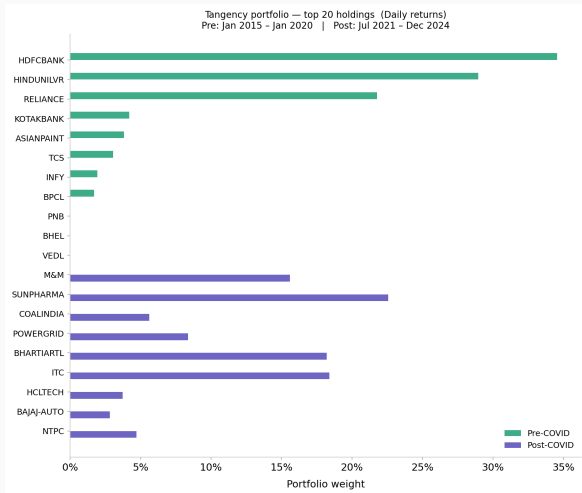
Why Returns Increased

- Sectoral rotation into high-growth areas (pharma, tech, metals, autos)
- FII inflows from China+1 strategy
- PLI schemes boosting industrials
- Persistent equity risk premium expansion

Why Volatility Increased (But Less Than Returns)

- Inflation uncertainty, geopolitical shocks, Fed tightening transmitted globally
- **But:** Correlation structure improved; sectoral dispersion was *diversifiable*
- SIP flows dampened daily volatility in large-caps

Tangency Portfolio: Sectoral Rotation



Pre-COVID top holdings

- HDFCBANK 34.5%
- HINDUNILVR 29.0%
- RELIANCE 21.8%

Post-COVID top holdings

- SUNPHARMA 22.6%
- ITC 18.4%
- BHARTIARTL 18.2%
- M&M 15.6%

Complete rotation out of **financials & FMCG** into **pharma, telecom, energy, autos**

Five Structural Drivers

1. **Monetary regime change** — RBI accommodation expanded equity risk premia; subsequent tightening added volatility without killing returns
2. **Sectoral recomposition** — Complete rotation from financials/FMCG to pharma/tech/metals/autos; winners won *bigger*
3. **Correlation decline** — Post-crisis recovery featured *lower* cross-sector correlations than pre-COVID; diversification improved
4. **China+1 FII flows** — Record foreign inflows improved liquidity, compressed spreads, reduced idiosyncratic volatility
5. **Retail participation surge** — SIP flows (8k → 20k cr/month) stabilized large-cap volatility; systematic buying absorbed dips

These are structural, not cyclical — the new frontier may persist.

Stock-Level: Biggest Winners

Ticker	Company	Pre	Post	Δ
VEDL	Vedanta	+8.0%	+51.2%	+43.2 pp
TATAPOWER	Tata Power	-4.9%	+48.7%	+53.5 pp
BHEL	Bharat Heavy Electricals	-20.7%	+46.8%	+67.5 pp
M&M	Mahindra & Mahindra	+0.4%	+43.0%	+42.6 pp
SUNPHARMA	Sun Pharmaceutical	-10.4%	+34.1%	+44.5 pp
HCLTECH	HCL Technologies	+8.6%	+32.4%	+23.8 pp

Industrials, metals, pharma, and IT led the post-COVID recovery.
BHEL and TATAPOWER flipped from worst to top performers

Stock-Level: Pre-COVID Stars That Faded

Ticker	Company	Pre	Post	Δ
KOTAKBANK	Kotak Mahindra Bank	+21.1%	+4.6%	−16.5 pp
HDFCBANK	HDFC Bank	+19.7%	+10.6%	−9.1 pp
HINDUNILVR	Hindustan Unilever	+17.9%	+7.9%	−10.0 pp
ASIANPAINT	Asian Paints	+17.8%	+9.4%	−8.4 pp
BPCL	BPCL	+23.5%	+15.3%	−8.2 pp

Financials and consumer staples — the pre-COVID defensives — underperformed significantly post-COVID.

These are also the stocks that *dominated the pre-COVID tangency portfolio*.

Limitations

- **Survivorship bias** — fixed 2015 universe excludes stocks that failed or were delisted; biases results upward
- **Long-only constraint** — rules out short selling; actual attainable frontier may be wider
- **Transaction costs & liquidity** — not modeled
- **Causal attribution** — the drivers discussed are correlational; isolating macro causes requires further regression analysis

Future Roadmap

1. **Jobson-Korkie test** — Test whether the Sharpe ratios of the pre- and post-COVID tangency portfolios are statistically different from each other, accounting for sampling uncertainty
2. **De Roon & Nijman spanning test** — Determine if the post-COVID frontier is truly outside the pre-COVID frontier, or whether it can be spanned by the original assets with different weights
3. **Frontier shift decomposition** — Decompose the shift into mean return changes vs. covariance structure changes; isolate whether improved Sharpe ratios come from higher expected returns or better diversification
4. **Covariance matrix estimation robustness** — Compare Ledoit-Wolf shrinkage with alternative estimators (POET, factor models, DCC-GARCH); assess sensitivity of frontier position to covariance estimation method
5. **Bootstrap confidence intervals** — Construct 95% confidence bands around both frontiers; quantify the probability that the observed gap is due to sampling noise versus a genuine structural shift
6. **Transaction cost modeling** — Estimate rebalancing costs for the tangency portfolio; assess whether the Sharpe improvement survives after accounting for bid-ask spreads and turnover

Thank you

Questions?